

Dimensionality Reduction using PCA and t-SNE on the Breast Cancer Dataset

1. Objective

The objective of this task is to understand and implement dimensionality reduction techniques using Principal Component Analysis (PCA) and t-distributed Stochastic Neighbor Embedding (t-SNE). These techniques help reduce the number of features in high-dimensional datasets while preserving important information. The task also aims to compare the performance and visualization capabilities of PCA and t-SNE using the Breast Cancer dataset.

2. Dataset Description

The Breast Cancer dataset was used for this task. It contains diagnostic measurements computed from digitized images of breast cell nuclei. The dataset is commonly used for binary classification problems.

Attribute	Value
Dataset	Breast Cancer Dataset
Number of Samples	569
Number of Features	30
Target Classes	Malignant (0), Benign (1)
Missing Values	None

3. Libraries Used

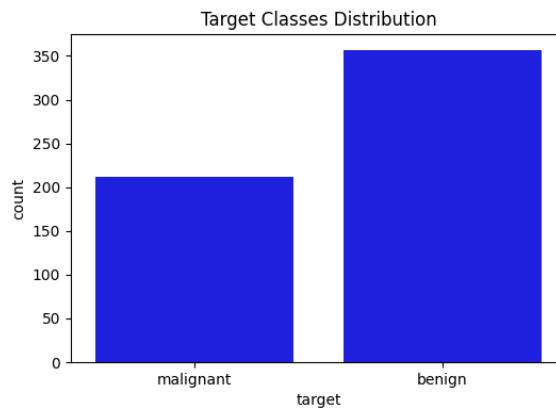
Library	Purpose
Pandas	Data manipulation
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical plots
Scikit-learn	PCA, t-SNE, preprocessing

4. Data Preprocessing

Before applying dimensionality reduction, the dataset was explored and preprocessed. The first few records were displayed, dataset information was examined, and missing values and duplicate records were checked. Since PCA is sensitive to feature scales, the numerical features were standardized using StandardScaler so that each feature had a mean of zero and a standard deviation of one.

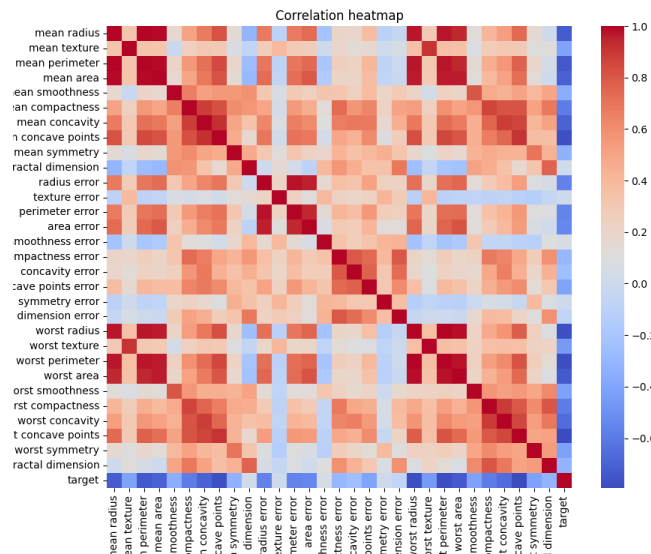
5. Exploratory Data Analysis (EDA)

Target Class Distribution



The target distribution plot shows the number of malignant and benign cases in the dataset. The dataset is reasonably balanced, making it suitable for analysis.

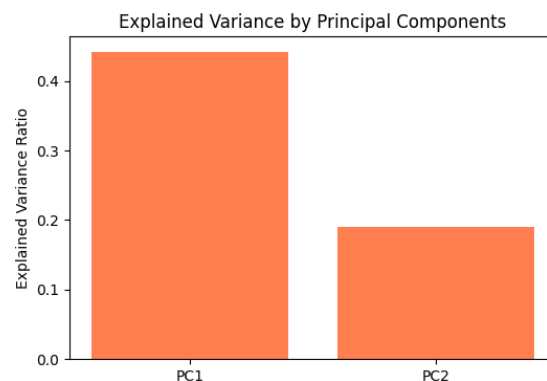
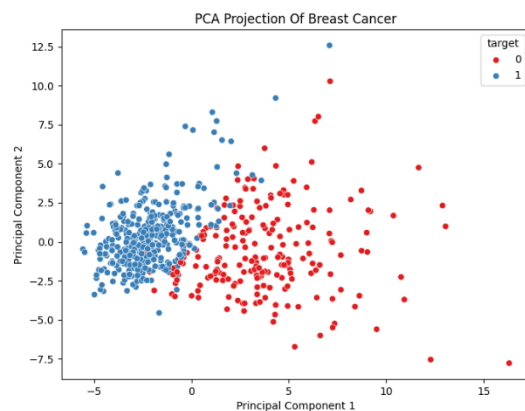
Correlation Heatmap



The correlation heatmap illustrates the relationships between different features. Several features show strong positive correlations, indicating redundancy in the dataset. This makes dimensionality reduction techniques like PCA particularly useful.

6. Principal Component Analysis (PCA)

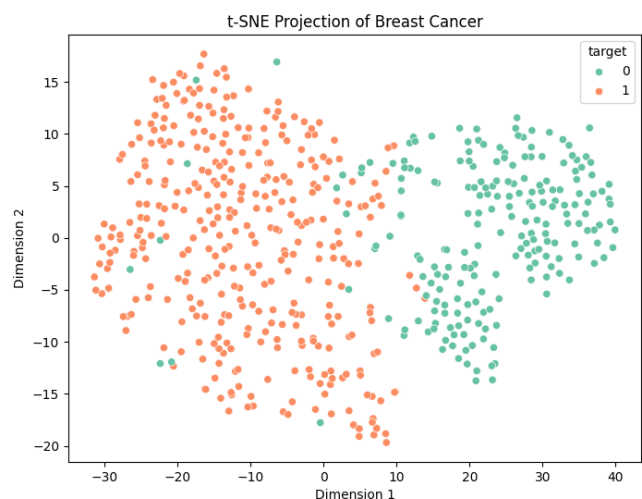
Principal Component Analysis (PCA) was applied after standardizing the dataset. The original 30 features were reduced to 2 principal components while retaining as much variance as possible. These principal components represent new features that summarize the original data.



The PCA scatter plot visualizes the data in two dimensions. The malignant and benign classes show noticeable separation, indicating that PCA successfully preserved important information despite reducing the number of features.

The explained variance plot shows how much variance is captured by each principal component. The first principal component retains the highest proportion of information, while the second contributes additional variance.

7. t-SNE



The t-SNE algorithm was applied to project the high-dimensional dataset into two dimensions while preserving local relationships among data points. Compared to PCA, the visualization produced by t-SNE shows a clearer separation between malignant and benign samples, making it highly effective for visualization purposes.

8. Comparison of PCA and t-SNE

Feature	PCA	t-SNE
Technique	Linear	Non-linear
Speed	Fast	Slow
Preserves	Global Variance	Local Structure
Best Use	Feature Reduction	Data Visualization
Computational Cost	Low	High

PCA is computationally efficient and suitable for reducing features before machine learning. In contrast, t-SNE is primarily used for visualization because it preserves local neighborhoods and often produces clearer clusters, although it requires more computational resources.

9. Conclusion

This task demonstrated the implementation of two popular dimensionality reduction techniques: PCA and t-SNE. PCA effectively reduced the number of features while retaining a significant amount of the dataset's variance, making it suitable for preprocessing and improving computational efficiency. t-SNE provided a more visually distinct representation of the data by preserving local structures, making it ideal for visualization. Both techniques successfully simplified the high-dimensional dataset while maintaining valuable information.